

Verdant Coil – Full Game Vision & Flow Document

This document consolidates all insights, flows, and designs discussed so far, enriched with details from the design documents and progress logs. It is intended as a **verbose reference** to give a clear picture of the game experience, player journey, and technical structure.

Concept Summary

Verdant Coil is an asynchronous, tile-based dungeon builder and explorer set in a surreal, biotech-inspired world. Players alternate between **building** organic maze-like dungeons (Coils) and **exploring** those created by others. The game's progression is powered by two resources:

- **Nutrient** – earned by exploring Coils. Spent on crawler upgrades and progression.
- **Sporeprint** – earned passively when other players complete your published Coils. Spent on unlocking advanced tiles, hazards, and higher biomass budgets.

The design pillars: - **Asynchronous creative ecosystem** – levels live on beyond their creator. - **Biotech world-building** – alien, organic, and alive. - **Progress through participation** – both playing and being played matter. - **Deterministic tactical depth** – puzzles with clear, testable rules. - **Accessible complexity** – simple to start, but rich to master.

The Big Picture Loop

Explore → Earn → Build → Publish → Get rewarded → Explore...

1. **Explore** a Coil in Explorer Mode. Reach the Heartroot to earn **Nutrient**.
2. **Earn** Nutrient → spent on crawler upgrades.
3. **Build** a new Coil in Builder Mode within a **biomass budget**. Place hazards, walls, pools, and markers.
4. **Publish** once validated. The Coil must be solvable by its creator.
5. **Get rewarded** when other players complete your Coil. You gain **Sporeprint**, which unlocks advanced tiles and higher biomass caps.
6. Loop back to **Explore**, stronger and with more tools to create.

This loop is asynchronous: you never play against others in real time, but your creations and completions fuel each other.

The Heartroot Hub (Main Scene)

The **Heartroot Hub** (Main.tscn) acts as the living menu and switchboard. It manages transitions, player data, and global state.

Functions:

- **Play (Explore)** – browse local or shared Coils; earn Nutrient by completing them.
- **Build (Creator)** – enter Builder Mode to design new Coils.
- **Upgrades** – spend Nutrient on crawler upgrades (toggles in MVP, trees later).
- **Library/Browser** – access saved drafts, published levels, and metadata.
- **Profile/Stats** – track personal stats: coils built, completions, high scores, play count.
- **Settings/Credits** – standard menus.

Layout Language:

- **Top bar = resources/tools.** Nutrient, Sporeprint, cog icon for settings.
- **Center grid = big choices.** Play, Build, Upgrades, Library.
- **Side panels = context.** Left: stats & activity. Right: drafts/published Coils.
- **Bottom = whisper UI.** Debug hints, small badges, dev indicators.

Visually, the hub is a **living biotech tree**: calm, pulsating, alien but serene.



Builder Mode (Creator Flow)

Goal: Craft a Coil under biomass constraints; validate it; then save/publish.

Layout (MVP)

- **Playfield:** CoilMap with four layers (Base, Walls, Hazards, Markers) + a Preview ghost tile.
- **Top bar:**
- Left: **Tile palette** (Flesh, Walls, Digest Walls, Acid, Sticky, Spawn, Heartroot, Eraser).
- Right: **Biomass counter** (current / cap, turns red when over cap).
- Actions: **Validate, Save, Publish.**
- **Bottom corners:** Undo (left), Dev badge (center), debug hints (right).
- **Status label:** Inline one-liner feedback.

Core Rules

- Must place **exactly one Spawn** and **one Heartroot**.
- Biomass cap enforced: no placement over limit.
- Illegal placement highlighted in red.
- Singletons enforced (placing a second Spawn/Heartroot replaces the first).

Validation Button (MVP)

Runs the following checks: 1. One Spawn exists. 2. One Heartroot exists. 3. Biomass $\leq$ cap. 4. **Path exists** between Spawn and Heartroot (flood-fill through walkable terrain).

Result modal: - **Green (success):** "Valid coil. Ready to Save/Publish." Options for **Playtest** and **Save**. - **Red (failure):** Bulleted list of unmet rules ("Missing Spawn," "Biomass over by 76"). Optional highlight flashes problem tiles.

Validation is modal (popup) to keep feedback explicit and focused.

Explorer Mode (Player Flow)

Goal: Navigate the Crawler to the Heartroot while surviving hazards.

Layout (MVP)

- **Top bar:** Resource counters (Nutrient, Sporeprint placeholder), upgrade toggles.
 - **Playfield:** `CoilMap` with same layered structure as Builder.
 - **Bottom corners:** Restart () and Menu ()
 - **Crawler gameplay:**
 - Moves deterministically, one tile per press.
 - Hazards resolve in order (acid damage, sticky slow, wall checks).
 - Win condition = reach Heartroot.
 - **Win moment:** “Heartroot Reached” ribbon; button to return to Hub.
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Session Stories

“I want to build.”

1. Hub → Builder.
2. Place tiles, hazards, markers.
3. Biomass updates live.
4. **Validate:** red prompts until fixed.
5. **Playtest:** instantly loads the Coil in Explorer Mode.
6. **Save/Publish:** save JSON locally (Phase-1.5). Published folder stub.
7. Back to Hub.

“I want to play.”

1. Hub → Level Browser.
 2. Select a Coil.
 3. Explore it in Explorer Mode.
 4. Reach Heartroot → earn Nutrient.
 5. Back to Hub → spend Nutrient on upgrades.
 6. Optionally → return to Builder with more tools unlocked.
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Tiles, Hazards & World Elements

From design docs [【129†Verdant Coil Design.pdf】](#) [【131†Verdant Game Elements.pdf】](#) [【132†Verdant Coil Overview.pdf】](#) :

- **Flesh Tile** – walkable base.
- **Digest Wall** – blocks movement unless Acid Sac is active.
- **Acid Pool** – damages crawler unless Hardened Skin.
- **Sticky Web** – slows crawler unless mobility upgrade.
- **Heartroot** – goal tile.
- **Pulse Gate, Trigger Node, Guard Nodule, Spine Shooter, Tendril Lash** – post-MVP expansion.

Hazards resolve in deterministic turn order: 1. Pre-check blocking (bounds, walkable, upgrade gating). 2. Move. 3. Apply hazard effects (acid damage, sticky delay). 4. Check for goal.

Resources & Progression

- **Nutrient** (earned by playing): spent on upgrades.
- **Sporeprint** (earned when others beat your Coils): spent on unlocking advanced tiles, hazards, biomass expansion.

Publishing rule: you must complete your own Coil in Explorer Mode before publishing.

Phased Roadmap (MVP → Expansion)

From roadmap [【130†Verdant Coil Plan.pdf】](#) :

- **Phase 1:** Core gameplay prototype.
 - 1.1 Crawler movement  [【139†1.1 progress log.txt】](#)
 - 1.2 Upgrade system  [【140†1.2 progress log.txt】](#)
 - 1.3 Tile system + hazards  [【142†1.3 progress log.txt】](#)
 - 1.4 Builder Mode (palette, biomass, validate, save/publish) 
 - 1.5 Save/Load JSON 
 - **Phase 2:** Greybox polish, undo/redo, QA tools, profile tracking.
 - **Phase 3:** Local sharing – level browser, ratings.
 - **Phase 4:** Online sync – accounts, anti-abuse, backend.
 - **Phase 5:** Meta systems – full economy, cosmetics, dashboards.
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Visual & UX Style

From UI guide [【132†Verdant Coil Overview.pdf】](#) : - Theme: **alien biotech** (deep teals, glowing greens, muted purples). - UI elements: **pill-shaped buttons**, translucent overlays, calm typography. - Layout rule: **top = tools/resources**, **center = playfield**, **bottom = whispers/debug**. - Effects: hazards pulse subtly, Heartroot glows, upgrades show via shaders (later polish).

Why This Flow Works

- **Ping-pong loop:** Build teaches you how to play; play gives you rewards to build better.
 - **Validation ensures fairness:** no unsolvable or troll levels.
 - **Short creative cycles:** paint → validate → playtest, all in seconds.
 - **Asynchronous growth:** your Coils live on and reward you even while offline.
 - **Scalable foundation:** MVP is simple, but architecture (Scene Tree, managers, TileMap metadata) supports long-term expansion [【134†Verdant Scene Architecture.pdf】](#) .
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Final Word

Verdant Coil is designed to feel like **growing puzzles inside a living organism**. Each mode is lean and purposeful: - **BuilderMode** = crafting with rules. - **ExplorerMode** = solving with upgrades. - **Heartroot Hub** = your living bridge between them.

Together, they create a game world where creativity and discovery loop endlessly, feeding into each other like the coils of an organism.